

synchronization, and configuration commands (via the ConfigBUS path). Two ProASIC FPGAs buffer the resets and JTAG signals (TMS, TCK and TDI) received from the GBTx chips for the remote reconfiguration of the Xilinx Kintex Ultrascale (KU) FPGAs. The two 20 nm planar featured KU FPGAs distribute clocks and configuration received from the GBTx to the MB, while reading slow integrator data and fast digitized data from two different gains of the PMTs. The xADCs of the KU FPGAs sample temperature, voltage stability and currents. Additionally, the KU FPGAs are connected to the Cs and xADC headers for extra digital and analog capabilities respectively, and communicate with each other for monitoring purposes via the so called CommBUS and GTHBUS. An over-current protection circuit triggers a power cycle in the case of the presence of high currents. Each KU FPGA sends data off-detector by means of two copies of GBT-FEC protected words through two 9.6 Gbps uplinks. The board radiation tolerance is achieved by the redundancy layers in the design, using the Xilinx Soft Error Management (SEM), Triple Mode Redundancy (TMR), GBT-FEC protocol, and GBTx ASICs.

3 Radiation requirements for the DB6

The radiation levels that the DB is expected to be exposed to were calculated from the radiation simulations performed by the ATLAS Collaboration to estimate the expected radiation levels for the HL-LHC era [9]. The most irradiated positions where the DB will be installed was used as a reference value for the radiation requirements. Approximately 108 Gy is the Total Ionizing Dose (TID) required for the design to be qualified, taking into consideration Safety Factors (SF) of 1.5 for simulation and 3.0 for lot variation in the components. The Non-Ionizing Energy Losses (NIEL) fluence required is for qualification is 13.16×10^{12} n/cm², including SFs of 1.5 for simulation, 1.3 when using a non-monoenergetic neutron source, and 3.0 to account for lot variation of the components. The expected fluence for Single Event Effects is 4.54×10^{11} h/cm², where a SF = 3.0 for simulation was included in the calculations. All the aforementioned requirement values have been put into place in accordance with the ATLAS Radiation Tolerant Electronics Proposed Guideline on COTS Lot to Lot Variation [10].

4 Radiation tests on the DB6

Four Daughterboards were produced and separately used to test for radiation tolerance, labelled in this document as DB6-0, DB6-1, DB6-2 and DB6-3.

Two separate tests were performed to cover tolerance to TID. The KU FPGA temperatures and all the currents corresponding to all the point of load regulators of the irradiated boards were monitored. Special attention was placed on the total of ten Coretek SFP+, the six KU FPGAs, the six Microsemi ProASIC3 FPGAs and the IS25LP128F reconfiguration FLASH memories. None of the components were damaged by the deposited TID: DB6-1 and DB6-2 were irradiated with Gamma radiation at ⁶⁰Co at the CERN CC60 facilities to 220 Gy at a rate of 3.37 Gy/h (figure 9 of ref. [11]) and up to 54 Gy at a rate of 0.33 Gy/h (figure 10 of ref. [11]), respectively. DB6-1 ran stable during the full test, however it showed an increase in the current measured in the 0.95 V power line of both sides of the board of at around 140 Gy. This was demonstrated to be correlated with the failure of the active components of the fan used to keep the KU FPGAs cool, leading to the increase